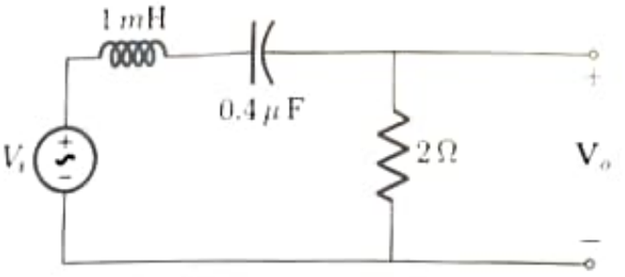
EE2015 (Electric Circuits and Networks) final exam

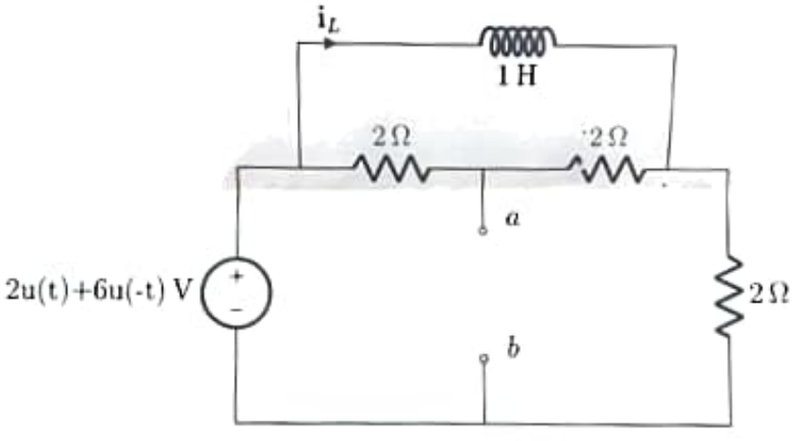
Marks: 60, Duration: 3 hrs

November 18, 2024

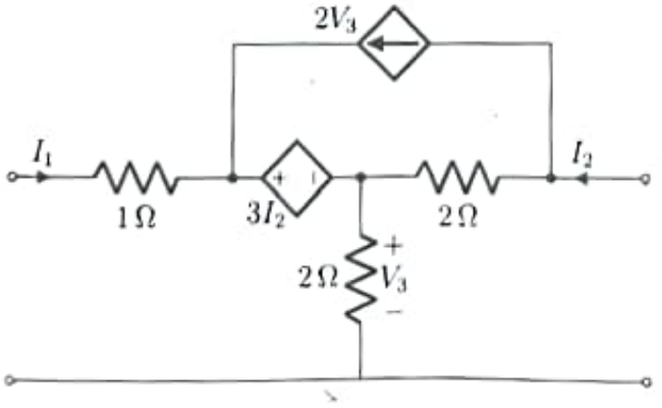
1. Consider the circuit shown on the right.
   (a) Find the magnitude of $\frac{\mathbf{V}_o(jw)}{\mathbf{V}_i(jw)}$
   (b) Find the resonant frequency (ω_0), Q-factor and bandwidth. (4)



2. Consider the circuit shown below.
   (a) Compute $i_L(0^-)$.
   (b) Obtain the s-domain Thevenin equivalent circuit at terminals $a-b$ for $t>0$
   (c) A 0.25F capacitor is connected at the terminals $a-b$. Using the Thevenin equivalent circuit obtained in part(b), compute $v_C(t) = v_{ab}(t)$. (6)



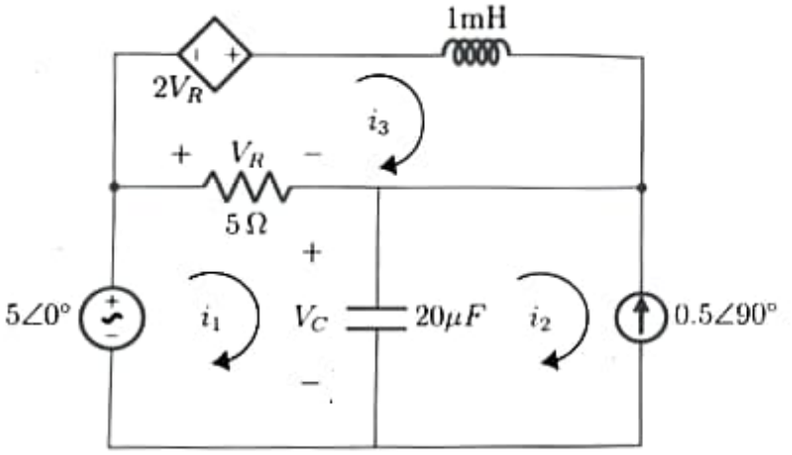
3. Find the g parameters of the following two-port network. (6)



4. For the circuit shown below, $\omega = 10000$ rad/s. Using mesh analysis,

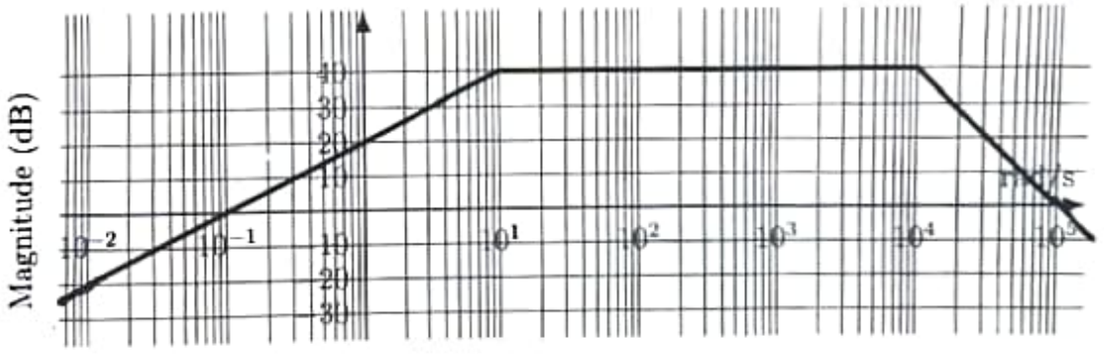
   (a) Obtain the circuit equations with I_1, I_2 and I_3 as the unknown variables. Put the equations in the matrix form. **All entries must be in the $a + jb$ format.**

   (b) Simplify the equations and determine $v_c(t)$. (4)



5. The magnitude Bode plot of a network is shown below. Assume that the poles and zeros are either on the imaginary axis or on the left-half of the s-plane.

   Derive the transfer function and draw its phase Bode plot. What will be the change in the transfer function if it is given that the phase at $\omega = 1000$ rad/s is $\angle 180°$? (6)



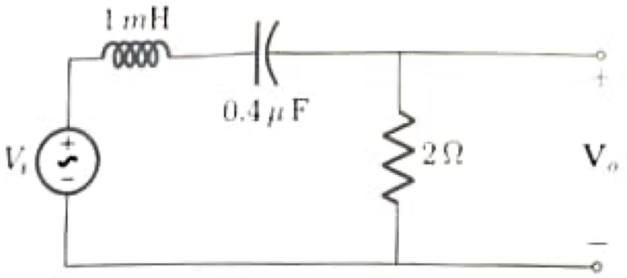
6. A balanced three-phase Δ-connected load (ABC/RYB) is supplied from a balanced three-phase source (abc/ryb) through a three-wire system with an impedance of $0.2 + j0.3\,\Omega$ in each line. The source-end line voltage, $\mathbf{V}_{ab} = 400\angle 30°$ V (rms) and load branch current, $\mathbf{I}_{BC} = 20\angle -36.87°$ A (rms).

   Find (i) the load-branch current, $\mathbf{I}_{AB}$ assuming positive phase sequence (ii) the line current, $\mathbf{I}_{aA}$ (iii) the load-branch voltage, $\mathbf{V}_{AB}$ (iv) power factor of the load, v) total active power delivered to the load (vi) total active power supplied by the source end and the power factor seen by the source and (vii) total instantaneous power seen at the source. (8)

1. Consider the circuit shown on the right.

(a) Find the magnitude of $\frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_i(j\omega)}$

(b) Find the resonant frequency (ω_0), Q-factor and bandwidth. (4)



(a)



$V_1 = I(R + j\omega L + \frac{1}{j\omega C})$

$V_o = IR$

$$\frac{V_o}{V_1} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

$$\left|\frac{V_o}{V_1}\right| = \frac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

(b) $\omega_0 L = 1/\omega_0 C$

$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} = 5 \times 10^4$ rad/s

$Q = \frac{\omega_0 L}{R} = 25$

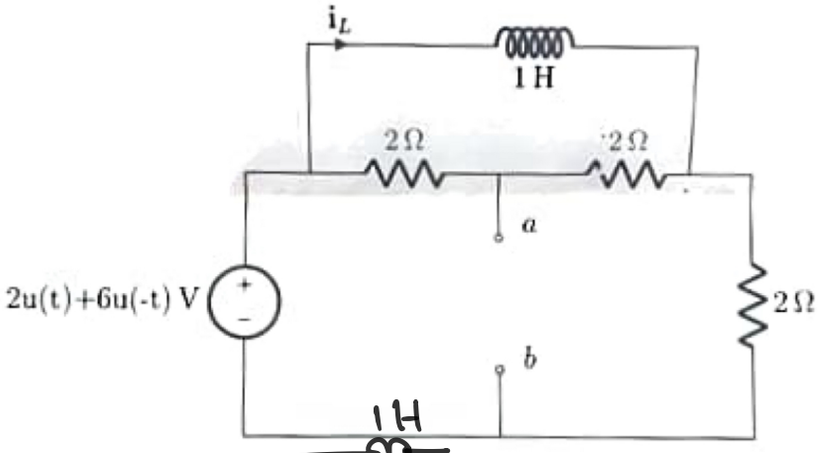
$BW = \frac{\omega_0}{Q} = 2000$ rad/s

2. Consider the circuit shown below.

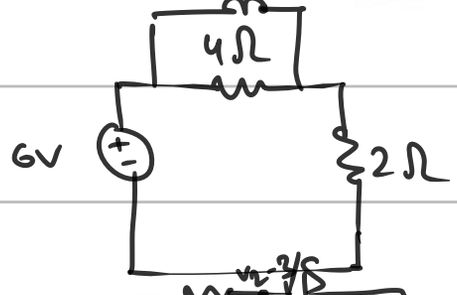
(a) Compute $i_L(0^-)$.

(b) Obtain the s-domain Thevenin equivalent circuit at terminals $a - b$ for $t > 0$

(c) A 0.25F capacitor is connected at the terminals $a-b$. Using the Thevenin equivalent circuit obtained in part(b), compute $v_C(t) = v_{ab}(t)$. (6)



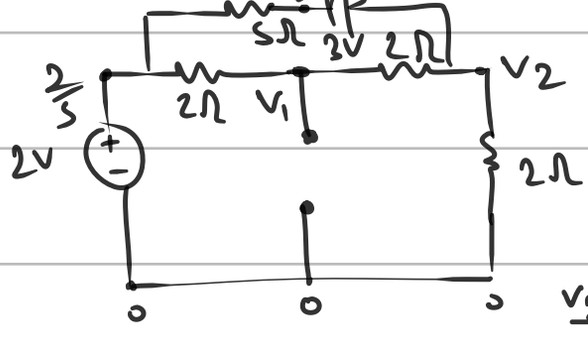
$t < 0 \Rightarrow$



at $t = 0^- \Rightarrow$ inductor acts as SC

$\Rightarrow i_L = \frac{6}{2} = 3A$

$t > 0 \Rightarrow$



$\frac{V_1 - 2}{5} = V_2 - V_1$

$\Rightarrow V_1 = \frac{V_2 + 2/5}{2}$

$\frac{V_2 - V_1}{2} + \frac{V_2 - 5}{5} = \frac{-V_2}{2}$

V_{th}:

$$V_1 = \frac{11}{3s+4} + \frac{1}{s} = \frac{14s+4}{s(3s+4)}$$

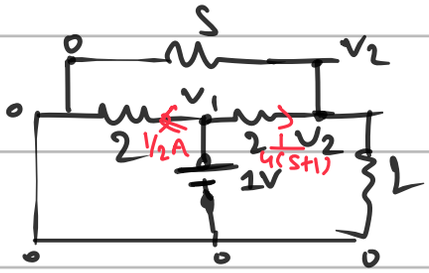
$$= \frac{V_2-2s}{4} + \frac{V_2-5}{s} = -\frac{V_2}{2}$$

$$\Rightarrow V_2\left(\tfrac{1}{4}+\tfrac{1}{s}+\tfrac{1}{2}\right) = \tfrac{5}{s}+\tfrac{1}{2s} = \frac{11}{2s}$$

$$\Rightarrow V_2\left(\frac{3s+4}{4s2}\right) = \frac{11}{2s}$$

$$\Rightarrow V_2 = \frac{22}{3s+4}$$

E_{th}:

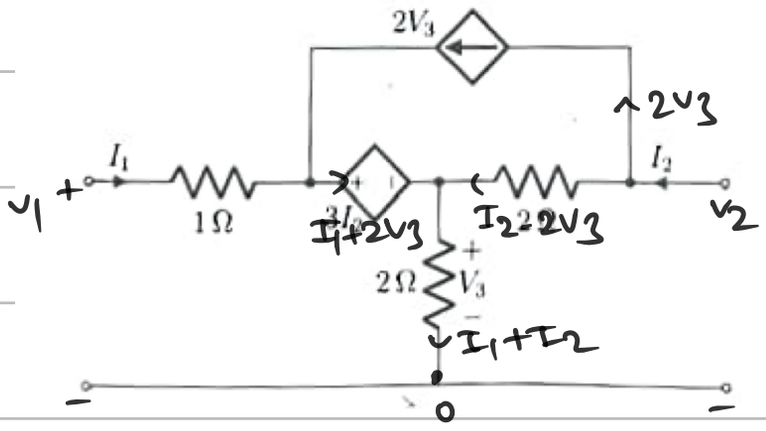


$$i = \frac{1}{2} + \frac{1}{4s+4} = \frac{4s+6}{2(4s+4)} = \frac{2s+3}{4s+4}$$

$$\frac{V_2}{s} + \frac{V_2-V_1}{2} = -\frac{V_2}{2} \qquad V_2\left(1+\tfrac{1}{s}\right) = \tfrac{1}{2} \Rightarrow V_2 = \frac{s}{2(s+1)}$$

$$\frac{V_2-V_1}{2} = \frac{\frac{s}{2(s+1)} - \frac{1}{2}}{2} = \frac{1}{4}\left[\frac{-1}{s+1}\right]$$

3. Find the g parameters of the following two-port network.



$$I_1 = g_{11}V_1 + g_{12}I_2$$

$$V_2 = g_{21}V_1 + g_{22}I_2$$

$$V_3 = 2(I_1+I_2)$$

$$V_1 = I_1 + 3I_2 + 2(I_1+I_2) = 3I_1 + 5I_2$$

$$V_2 = \left[I_2 - 4(I_1+I_2)\right]2 + 2(I_1+I_2) = -6I_1 - 4I_2$$

$$I_1 = \frac{V_1}{3} - \frac{5I_2}{3}$$

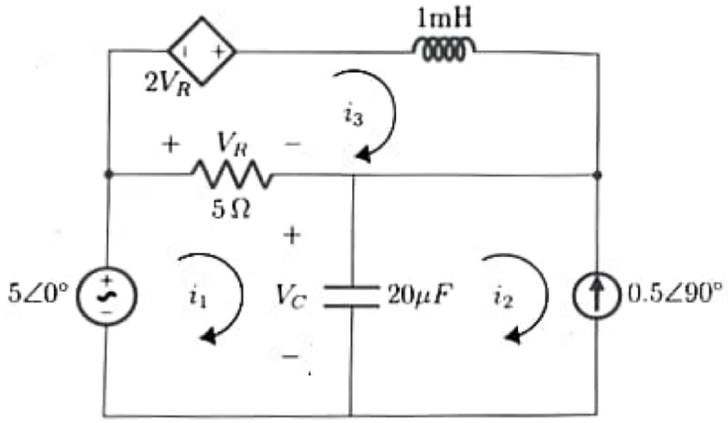
$$V_2 = -6\left(\frac{V_1}{3} - \frac{5I_2}{3}\right) - 4I_2 = -2V_1 + 6I_2$$

$$\begin{bmatrix} 1/3 & -5/3 \\ -2 & 6 \end{bmatrix}$$

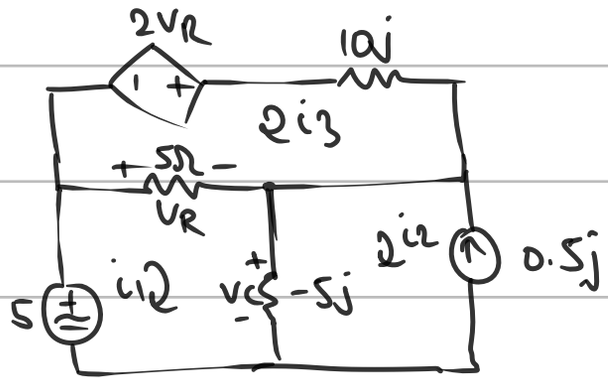
4. For the circuit shown below, $\omega = 10000$ rad/s. Using mesh analysis,

(a) Obtain the circuit equations with I_1, I_2 and I_3 as the unknown variables. Put the equations in the matrix form. **All entries must be in the $a + jb$ format.**

(b) Simplify the equations and determine $v_c(t)$. (4)



$$\frac{1}{j \cdot 10^4 \cdot 2 \times 10^{-5}}$$



$$5 = 5(i_1 - i_3) + -5j(i_1 - i_2)$$

$$0.5j = -i_2$$

$$3(5)(i_1 - i_2) = 10j\, i_3$$

$$\begin{bmatrix} 5 \\ -0.5j \\ 0 \end{bmatrix} = \begin{bmatrix} 5-5j & 5j & -5 \\ 0 & 1 & 0 \\ 15 & 0 & -15+10j \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}$$

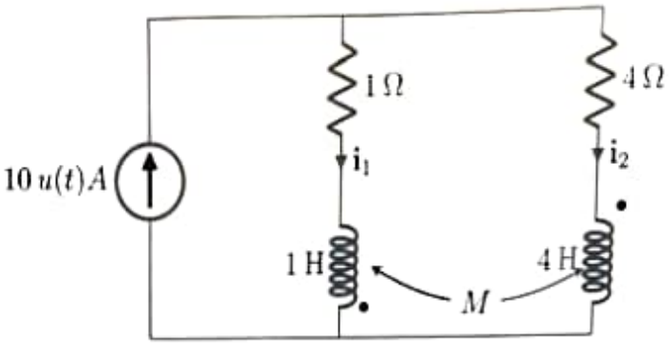
$$1 = (1-j)\, i_1 + j\, i_2 - i_3 \qquad i_2 = -0.5j$$

$$3i_1 = (3+2j)\, i_3 \qquad \boxed{v_c = \frac{2(1+j)}{2-j} \times -5j = \frac{10(1-j)}{(2-j)}}$$

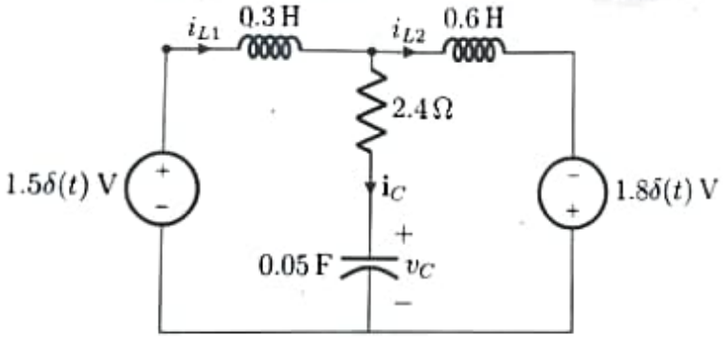
$$1 = (1-j)\, i_1 + 0.5 - \left(\frac{3}{3+2j}\right) i_1 \Rightarrow 0.5 = i_1 \left[\frac{5-j-3}{3+2j}\right]$$

$$i_1 = \frac{3+2j}{2(2-j)} \qquad i_1 - i_2 = \frac{3+2j}{2(2-j)} + \frac{j}{2} = \frac{4+4j}{2(2-j)}$$

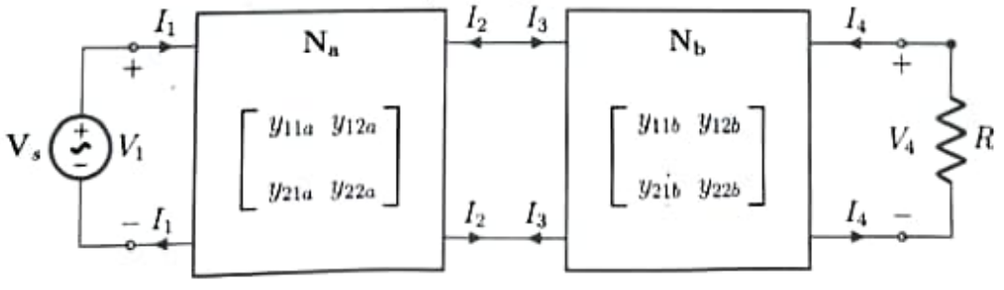
7. Consider the circuit shown below. Using s-domain analysis, find and plot $i_1(t)$ for $t >= 0$ for two cases of coefficient of coupling of mutual inductance, k=0 and k=1. Assume zero initial conditions. (6)



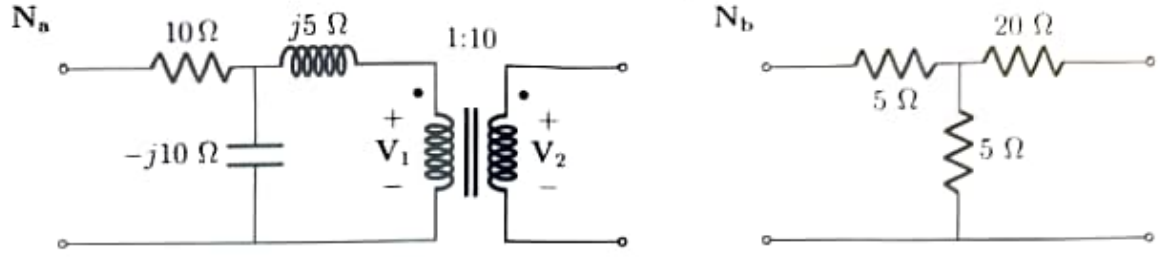
8. Consider the circuit shown below.
   (a) Using time domain analysis, find $i_{L1}(0^+)$, $i_{L2}(0^+)$, $i_C(0^+)$ and $v_C(0^+)$.
   (b) Draw the s-domain circuit for $t > 0$.
   (c) Compute the damping factor (α) and resonant frequency (ω_0) of the circuit. Is the circuit over/critically/under damped?.
   (d) Using the parameters obtained in part (c), write the general form of the time-domain solution for $v_C(t)$, $t > 0$. Use the initial conditions derived in part (a) to determine the constants in the general form of the solution. (6)



9. (a) For the network shown below, compute $\frac{V_4}{V_s}$ in terms of the y parameters of N_a and N_b.

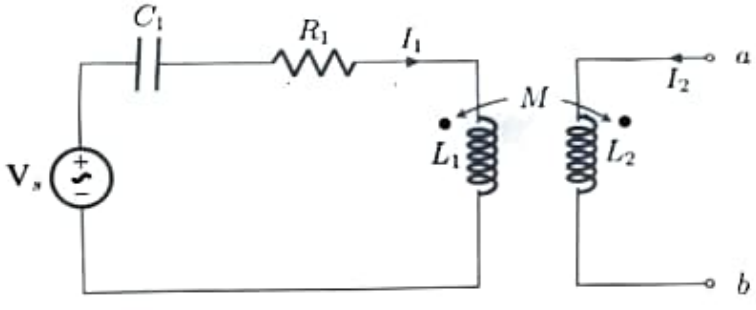


(b) Compute the y parameters of the two networks shown below.



(c) If $R = 100\ \Omega$, compute $\frac{V_4}{V_s}$ using the results of part (a). (8)

10. In the circuit shown below, $C_1 = 0.1$ F, $R_1 = 4\ \Omega$, $M = 0.5$ H, $L_1 = 1$ H, $L_2 = 2$ H and $V_s = \cos 2t$ V. **All angles should be expressed in degrees.**



(a) Find V_{oc}, I_{sc} and Z_{th} at the terminals $a - b$.

(b) Choose the load connected to $a - b$ for maximum power transfer and compute the average power transferred to it. (6)